rfsurrogates: share the solver time, share the models

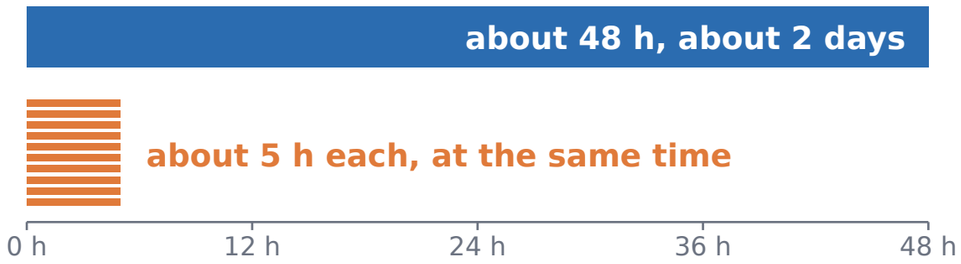
A surrogate model needs a few hundred full-wave solves. Contributors share the solves. The models and the data are open.

Solver time is the bottleneck

One full-wave solve: about 11 min on a desktop. One surrogate query: 4.5 ms.

One desktop
261 solves x 11 min

10 contributors
about 26 solves each



Open models, open data

An open alternative to closed commercial models.

	rfsurrogates	Closed models
Code, weights	open, Apache-2.0	closed
Data	open, CC-BY-4.0	not published
Who adds data	anyone	the owner

Family 1: 261 Palace solves. Held-out passband |S21| error:
0.52 dB for the trained model, 2.86 dB for the textbook prior alone.

1 Contributor solves designs

on their own machine, to their own budget

```
python tools/contribute.py go \
    --family edge_coupled_bpf --n 20
```

First run Draws designs from the parameter box. Opens a claim issue on GitHub.

Claim gate: the maintainer accepts

Second run Builds each case.

Geometry gate: rejects a wrong layout

Solves each case on this machine.
Checks each record.
Opens one pull request.

Solver Palace: runs automatically.
HFSS, CST or openEMS: the tool exports the geometry. You solve it. The tool reads the Touchstone file back.

2 CI verifier

on the pull request

GitHub Actions runs the checker again on every record.

PASS

NEEDS REVIEW

FAIL

3 Maintainer merges

records are pooled

The records go to the Hugging Face dataset

`rfsurrogates/samples`

and to the data index. Each record names its contributor.

4 Retrain, publish

open weights

Retrain when a family has about 25 % more records.

New weights go to Hugging Face.

Anyone can use them:

- browser demo
- inverse design
- download

more contributors → **more records** → **better models** → **more device families**